Academy 4Digital Scenario Demonstration - Mock

Level 3 Data Technician

Assessment method: Scenario Demonstration with Questioning

Duration: 90 minutes (+10% if required)
Assessment type: Open book – use of internet allowed for guidance but not to complete tasks

Scenario Description (≤750 words)

You have recently joined the UK Olympic Committee's Data Analysis Team. The committee has tasked you with preparing a data-driven report comparing medal results from the Tokyo 2021 Olympic Games across countries. The aim is to understand performance patterns, ensure the accuracy of the datasets, and present findings in a clear, reliable format for senior stakeholders.

You have been given three datasets:

- Tokyo Medals 2021 (CSV) – medal counts by country.
- Country Details (CSV) – metadata including demographics and economic indicators.
- Country Population (TXT) – 2020 population figures.

As a Data Technician, your responsibilities include:

- Gather and combine datasets into a structured format.
- Check for consistency, duplicates, anomalies.
- Apply validation and repeatable methods.
- Analyse results and identify trends.
- Present findings clearly in tables, charts, and a short report.

Outputs will inform future funding and participation decisions. Accuracy and validation are essential.

Annexes

- **Dataset Annex 1:** Tokyo Medals 2021.csv (medal counts)
- **Dataset Annex 2:** Country Details.csv (country metadata)
- **Dataset Annex 3:** Country Population.txt (population figures)
- **Documentation Annex 4:** Organisational requirement – "Data outputs must include a validation record and clear explanation of methods used."

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Tasks

Task 1 – Data Gathering and Preparation

- Import datasets into a single Excel workbook with separate worksheets. (K4, K5, K6, S1, S2, S3)
- Audit datasets: record column names, formats, units, and inconsistencies. Summarise in a short report. (S3)

Task 2 – Data Combination and Transformation

- Use lookup/linking techniques to combine Country Population and Country Details. (K13, S4, S16)
- Link the combined dataset with the Olympic Medals dataset. (S5, S16)
- Resolve inconsistencies (e.g., country names). Record corrections. (K15, S6, S16)
- Apply filters to retain relevant fields. Justify choices in the report. (K15, S3)

Task 3 – Data Validation and Quality Checks

- Identify and correct common issues (duplicates, missing values, misalignments). (K14, S6, S7)
- Apply at least one repeatable validation method (formula, macro). (K14, S7)
- Demonstrate structured instructions or prompt-based techniques for transformation. (K26, S16)

Task 4 – Data Analysis and Presentation

- Identify top 5 and bottom 5 countries by chosen medal indicators. (K16, S5, S3)
- Create labelled graphs/tables to illustrate trends. (K16, S5, S3)
- Calculate medals per capita and include in analysis. (K16, S5)
- Provide short written analysis in the Summary Report. (S3)

Task 5 – Documentation and Output

- Produce a clear Summary Report including audit, validation, filtering, analysis. (S3)
- Reference organisational requirements. (S3)

Assessor Questioning (minimum 5)

The assessor will ask at least 5 questions during natural breaks and at the end. Example questions:

1. How did you check for and resolve inconsistencies across the datasets? (K15, S6)
2. Can you explain why you chose to filter specific columns, and what risks might exist if you included all data? (K15)
3. What methods could you use to automate or repeat parts of your analysis to improve efficiency? (K14, S7)

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4. How might prompt engineering or structured instructions support more effective data transformation in your task? (K26, S16)
5. How would you communicate these findings differently to a technical audience versus a non-technical stakeholder? (K11, K12, S8)

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